

U SPArC Style tutorial

This document is a written version of the videos in the U SPArC Style tutorial.¹

When should you focus on style?

First think about the content and structure. Focus on style when you are almost ready to hand in (a section of) the text. You can also try to keep the style a bit in mind while writing, so you don't have to change everything at the end.

What is U SPArC?

U SPArC is a tutorial that teaches you how to write in an academic and clear style. It covers five style principles:

- 1) Understandable information load
- 2) Strong order
- 3) Precise
- 4) Appropriate register
- 5) Concise

Try to apply these principles to your text when editing, to make the style of your text academic and clear.



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Principle 1: Understandable information load

Sentences should not be overly **long or complicated**. Consider how much your intended reader can process. Avoid stringing sentences together into one very long sentence, as in example 1a. Instead, you can split up the sentence, as in example 1b.

1a. We developed a model relating urban density to social interaction, and we propose that density positively affects psychological health, but we also found that urban density relates to physical health.



1b. We developed a model relating urban density to social interaction. We propose that density positively affects psychological health. However, urban density also seems to relate to physical health.



You also do not want to end up writing only very **short sentences**. There is no rule in terms of a particular number of words, but you may agree that the ones in 1c are too short.

1c. We developed a model. It relates urban density to social interaction. We propose the following. Density positively affects psychological health. However, urban density also seems to relate to physical health.



Sentence 2a is **too long**. It contains several comma's and instances of the word *and*, which may be good places to start new sentences. Sentence 2b has an understandable information load.

2a. The WIP module is being developed as an open source project, with the intention of showcasing the new technique and serving as a proof-of-concept and is being designed by a large team of engineers, which is based in Delft.



2b. The WIP module is being developed as an open source project. It is meant to showcase the new technique and serve as a proof-of-concept. The module is being designed by a large team of engineers, which is based in Delft.



How much your reader can process also depends on the content. If the content is very **difficult or new** for your reader, try writing slightly shorter sentences. Compare 3a and 3b.

3a. When the machine was running on petrol, the carbon dioxide emissions were higher, which suggests improved mixing and less cylinder-to-cylinder variation in contrast to the findings described in Section 3.



3b. When the machine was running on petrol, the carbon dioxide emissions were higher. This suggests improved mixing and less cylinder-to-cylinder variation. This contrasts with the findings described in Section 3.



So when deciding on whether to split, think about your intended reader. Sentence 4 contains quite a few words, but it can easily be processed. You can therefore leave it as it is.

4. The percentage of natural gas has slightly increased (30% to 34%), that of oil has slightly decreased (50% to 46%), and that of coal too (18% to circa 10%).



Example 5 sits on the fence and you decide for yourself with your reader in mind.

5a. As power produced by a wind turbine is proportional to the cube of the wind speed, it is common to calculate the energy generation potential of a district using the average wind speed data. ?

5b. Power produced by a wind turbine is proportional to the wind speed. It is therefore common to calculate the energy generation potential of a district using the average wind speed data. ?

Common mistake 1: If you decide to split a sentence, make sure to still write **complete sentences**. In 6a, the sentence *an MVP and a V1* is not complete.

6a. The project produced two major releases. An MVP and a V1. Both stayed within budget and met the expectations of the university and the client. ✗

6b. The project produced two major releases, an MVP and a V1. Both stayed within budget and met the expectations of the university and the client. ✓

Common mistake 2: Use **this** or **these** at the beginning of a sentence. Use **which** with a comma.

7a. There are certain failure modes under which the components could fail, which have to be taken into account during the design phase. ✓

7b. There are certain failure modes under which the components could fail. These have to be taken into account during the design phase. ✓

Common mistake 3: Do not leave out a **linking word** when splitting a sentence ('as' in 8a). You can replace it by 'therefore' in the split version of the sentence.

8a. As power produced by a wind turbine is proportional to the cube of the wind speed, it is common to calculate the energy generation potential of a district using the average wind speed data.

8b. Power produced by a wind turbine is proportional to the wind speed. It is common to calculate the energy generation potential of a district using the average wind speed data. ✗

8c. Power produced by a wind turbine is proportional to the wind speed. It is therefore common to calculate the energy generation potential of a district using the average wind speed data. ✓

Summary Principle 1: Understandable information load

Consider a sentence's information load. Each sentence is a chunk and you don't want to overload your reader. A mix of longer and shorter sentences is ideal.

Principle 2: Strong order

We give three suggestions for a clear order within a sentence:

- 1) Look at the position of the main verb and try to place it early in the sentence.
- 2) Do not overdo your use of the passive.
- 3) Keep together what belongs together

1) Look at the position of the main verb and try to place it early in the sentence.

Sentences that make you wait for the verb are harder to read. In sentences where the verb appears late (1a), move the verb forwards (1b). The result is a sentence with the order Subject (*the group*) – Verb (*investigated*) – Rest.

1a. The influence of the robot's material on its performance was also investigated.



1b. In addition, the group investigated the influence of the robot's material on its performance.



In sentences 2a en 2b, the subjects are italicized and the verbs are underlined.

2a. To inform our choices for the application's features, such that the app would help the users gain or improve self-regulated learning skills, *a literature review* needed to be done.



2b. *We* conducted a literature review to inform our choices for the application's features, such that the app would help the users gain or improve self-regulated learning skills.



2) Do not overdo your use of the passive.

If you use the passive too much as in 3a, your text will not be easy to read. Avoid the passive by making something else the subject of the sentence (*the experiment* in 3b).

3a. A quasi-static setup was used in the experiment, yielding six samples. Two types of fatigue test were performed to measure the relevant lengths and each curve and slope was analyzed. The steepness of the slopes was found to correlate with length. Furthermore, plateaus were observed which depended on repeatability. As a result, it is recommended to improve camera quality in future research and it is advised to add a quasi-static counterpart.



3b. *The experiment* used a quasi-static setup, yielding six samples. We performed two types of fatigue test to measure the relevant lengths and subsequently analyzed each curve and slope. The results show that the steepness of the slopes correlates with length. Furthermore, we noticed plateaus which depended on repeatability. As a result, we recommend improving camera quality in future research and adding a quasi-static counterpart.



You do not need to use the passive to discuss a figure as in 4a. You can make the figure the subject of the sentence, as in 4b. This again gives the order Subject – Verb – Rest.

4a. It can be seen from Figure 2.1 that the manufacturing procedure in high temperature conditions differs from that under lower temperatures.



4b. Figure 2.1 shows/ demonstrates/ illustrates that the manufacturing procedure in high temperature conditions differs from that under lower temperatures.



Likewise, you can start a sentence with 'this report describes' or 'this chapter discusses', making the report or chapter the topic of the sentence, as in 5b.

5a. In the next sections, background on quantum computing and quantum bits is given (Section 2.1) and the advantages of quantum computers are shown (Section 2.2).



5b. The next two sections provide background on quantum computing and quantum bits (Section 2.1) and lay out the advantages of quantum computers (Section 2.2).



3) Keep together what belongs together.

Sentence 6a aims to say 'the Structural index is used for benchmarking', but this is interrupted by other information. In 6b, the interrupting information is placed in a separate sentence.

6a. The Structural index is computed as the sum of three indicators, the number of horizontal, vertical, and speed interactions between aircrafts in a designated volume, and is used for ANSP benchmarking [2].



6b. The Structural index is used for ANSP benchmarking [2]. It is computed as the sum of three indicators: the number of horizontal, vertical, and speed interactions between aircrafts in a designated volume.



You can also move the interrupting information to the end of the sentence, as in 7b.

7a. The experiments used values that did not include a typical temperature during flight, namely temperatures ranging from 0 °C, the freezing point of ice, to -25 °C.



7b. The experiments used temperature values ranging from 0 °C, the freezing point of ice, to -25 °C, and thus did not use a typical temperature during flight.



Summary Principle 2: Strong order

1) Place the verb early in the sentence.

2) If you want to use the passive, first try another way, such as making something else the subject of the sentence.

3) Keep together what belongs together.

This will create strong sentences with the order Subject – Verb – Rest. A final tip is to read a paragraph out loud. If it doesn't sound good, maybe you can vary more in the structure of the sentences.

Principle 3: Precise

Precision is key in a report, and even more so in a scientific paper. You should make sure that what you write is completely clear and unambiguous.

Pay attention to **words that refer** to something previously mentioned in the text. Common ones are *this*, *these*, *that* and *those*, *it* and *they*, and *which*. The reader should not have to wonder to what exactly those words refer. In 1b, the word *these* refers to *the steps of the Storm approach*, but in 1a, the previous sentence does not mention any steps. You should always be able to identify the exact words that a word such as *these* refers to. Those other words should also stand nearby, preferably in the previous sentence.

1a. The research has been conducted according to the described approach. All participants followed these steps.



1b. The research has been conducted according to the steps of the Storm approach (see Chapter 3). All participants followed these steps.



We give four suggestions to clarify reference words:

- 1) **Rephrase** the words to which the referring word (here: *these*) refers back (1a-b above)
- 2) Replace the referring word (here: *which*) with a **fuller description** (2a-b)

2a. Hydro power in Eastern Europe has historically been centred around the river Danube, which is the largest power source in that area.



2b. Hydro power in Eastern Europe has historically been centred around the river Danube. Water power is the largest power source in that area.



- 3) Use a **summary noun** to refer to a larger idea (*finding* in 3a-b and *two advantages* in 4a-b)

3a. Recent research [2] suggests that 5 of the 330 known toxic chemicals are responsible for over 60% of air pollution. This differs from the results of [3], who conclude that air pollution has a large variety of sources.



3b. Recent research [2] suggests that 5 of the 330 known toxic chemicals are responsible for over 60% of air pollution. This finding differs from the results of [3], who conclude that air pollution has a large variety of sources.



4a. The client's spokesperson stated that the laptops are easier to configure than the desktop computers, the laptops are sold better, but the desktops are produced faster. This is why Menco computers is planning to increase their production of laptops.



4b. The client's spokesperson stated that the laptops are easier to configure than the desktop computers, the laptops are sold better, but the desktops are produced faster. Still, the two advantages have made Menco computers decide to increase their production of laptops.



- 4) Do not refer to something in a **heading**, but repeat the words in the title (5a-b)

5a.
3.2 BuienRadar
The next step in the design process was to compare the prototype to this widely used Dutch weather application.



5b.
3.2 BuienRadar
The next step in the design process was to compare the prototype to BuienRadar, a widely used Dutch weather application.



Aside from referring clearly, you also want to use precise words. Avoid vague words such as *several*, *relatively* and *many*. Try to replace them with **numbers or precise descriptions**, as in 6b.

6a. Jones and Wilson tested their design with many participants and conclude that it is relatively fast in several different weather conditions.



6b. Jones and Wilson tested their design with 120 participants. They conclude that it is the fastest of the three designs, both in sunny and rainy weather conditions.



For the same reason, avoid words such as *etcetera* and *and so forth*. These are also **vague**.

7. The decision to use aluminum will affect the weight, production time, costs, etc.



In addition, try to use **the most specific word** that expresses what you want to say. In 8a, *says* is a general word, while 8b shows that there are plenty of specific alternatives.

8a. Recent work by Carlson [6] says that...



8b. Recent work by Carlson [6] analyses/introduces/lays out/describes/argues/demonstrates...



If you see general words such as *looks at* in 9a, decide what you want to say and choose a specific alternative. If you have trouble finding the right word, a free online thesaurus can give you inspiration (<https://www.thesaurus.com>).

9a. This section looks at the design specifications.



9b. This section considers/analyses/explores/examines/reviews the design specifications.



Summary Principle 3: Precise

Academic writing should be precise. Try to refer clearly and to use precise words.

- 1) If you use a referring word such as *these* or *which*, you should be able to identify the exact words in the previous sentence to which it refers back.
- 2) Check for vague words such as *several*, and use the most specific word for what you want to say.

Principle 4: Appropriate register

Appropriate register means that you should choose words according to (a) the **type of text** you are writing, such as a report or scientific paper, and (b) **your intended readers**, such as a company's management or scientists working in a certain field. The style of a scientific paper is usually a bit more formal than that of a report.

Try to avoid words that are **too informal**.

1. INFORMAL	WRITE INSTEAD
like (the costs)	such as (the costs)
a lot of	much/many
(a) big (increase) 	(a) substantial/significant (increase) 
get (results)	obtain (results)

Also avoid words that are **too difficult and unusual** for your intended readers, as in 2a, because they make a text unclear and can sound pretentious.

2a. This *methodology* was used to calculate the effect of the *usage of acetylsalicylic acid*. 

2b. This *method* was used to calculate the effect of the *use of aspirin*. 

It is considered informal to use **contractions** such as *isn't* in 3a and to start a sentence with **and**, **but**, **so**, **also** or **yet**.²

3a. This issue *isn't* explained in detail. But *it'll* be covered more fully in Chapter 3. 

3b. This issue *is not* explained in detail, but *it will* be covered more fully in Chapter 3. 

Do not use **you** or **your**, as in 4a. This is definitely too informal.

4a. As *you* can see in Figure 2, ... 

4b. Figure 2 illustrates... 

It is also informal to use a **direct question**, meaning a question with a question mark.

5a. Previous research [3, 4] suggests that the feedback in simulators is a key aspect of training. However, is this always the case? 

5b. Previous research [3, 4] suggests that the feedback in simulators is a key aspect of training. Our study investigates whether this also applies under non-standard weather conditions. 

² Example 3 combines examples 3 and 4 in the video.

Verbs that consist of several words, such as *look into* or *come up with*, are sometimes more informal than **one-word alternatives**. A thesaurus can help you find alternative words.

6. MULTIPLE WORDS	ONE-WORD ALTERNATIVE
look into	explore/consider
come up with	develop/generate

Use **concrete words** as much as possible, especially if your topic is abstract. It is not the case that academic writing should be difficult to read. If relevant, it is good to give an example.

7a. Osmotic tolerance is the ability of an organism to withstand inhibition of growth immediately following salinity stress. ❌

7b. Osmotic tolerance is the ability of plants to keep growing immediately following a high salt concentration around the roots. ✅

Appropriate language in a paper or report is also **objective language**. The word *enormous* in 8a has an emotional feel. 8b shows some objective alternatives.

8a. Compressed sensing has enormous potential to reduce operational time for MRI. ❌

8b. Compressed sensing has vast/a great deal of/considerable potential to reduce operational time for MRI. ✅

Academic writing is not about **giving an opinion**, as in 9a. Instead, just describe the results (9b). Be factual: only say that you provide a *thorough* or *detailed* account of something if your account is indeed thorough or detailed.

9a. We got some encouraging results. ❌

9b. This section describes the results, first focusing on standard conditions and then on three non-standard scenarios. ✅

Using **we** used to be unacceptable, but this is changing. Consult a lecturer from your program or a few published papers in your field to decide whether you can use *we*. You can use *we* to refer to yourselves as the writers (10) or to describe something important that *you* did, such as making a decision (11). If you use *we* too much, it may sound unprofessional.

10. We first explain the rationale behind this set-up, before describing the outcomes in the next section. ✅

11. We decided to exclude those data for the following reasons. ✅

Summary Principle 4: Appropriate register

Avoid words that are informal or over-formal for the type of text and the intended reader. Try to use language that is a bit formal, concrete, and factual and objective.

Principle 5: Concise

Concise means: 'use the minimum number of words needed to say what you want to say'. You need to decide if the words are redundant. Words that do not add anything to a sentence should be removed, because they make the sentence unnecessarily long or unclear.

Check for places where you say that **something is important or necessary**. In 1a, the phrase *it is important to thoroughly explain that* is basically redundant, so it should be left out. If you really want to say that your research is crucial, make sure to explain why.

1a. It is important to thoroughly explain that the citizens' privacy is threatened by media monopolies.



1b. The citizens' privacy is threatened by media monopolies.



1c. Better: Media monopolies threaten citizens' privacy.



Consider whether **the reader needs the information** that you might be about to remove, such as the struck-through text in 2a. If you think that the text is helpful, you should leave it in.

2a. ~~However, a more recent report [4] shows that~~ the increase in aircraft movements is solely due to a rise in the number of cross-country flights.



2b. However, a more recent report [4] shows that the increase in aircraft movements is solely due to a rise in the number of cross-country flights.



3b is more concise than 3a. However, be careful with removing **linking words** such as *nevertheless*, which explain a link to the reader.

3a. Nevertheless, most approaches at the present time focus on only one type of flight, namely commercial air traffic flights.



3b. Most current approaches only address commercial air traffic flights.



Still, you can also overdo linking words (4a). In 4b, *however* is put in the middle of the sentence for variation.

4a. Nevertheless, most current approaches focus on commercial air traffic flights. ~~Moreover~~, most recent studies consider both local and cross-country flights [3, 4]. ~~In fact~~, both types of flight carry increasing numbers of passengers [5].



4b. What has received limited attention, however, is the resulting fuel consumption.



If a sentence starts with **there is** or **there are**, this is usually redundant. You can turn 5a into 5b by removing *there is* and moving *engineers* to the beginning of the sentence.

5a. There is increasing interest among engineers in this new technique.



5b. Engineers are increasingly interested in this new technique.



Just tell the reader what you did (6b), instead of saying that something can be done (6a). 6c is actually clearest.

6a. The pressure distribution can be examined by plotting the pressure coefficients for ten slices of the wing.



6b. The pressure distribution was examined by plotting the pressure coefficients for ten slices of the wing.



6c. Better: To examine the pressure distribution, we plotted the pressure coefficients for ten slices of the wing.



In 7a, you do not need to say that a careful design is needed: just explain the design.

7a. Careful design of the MOSFET experiment is needed in order to increase oxide reliability.



7b. Materials with high relative permittivity were used in order to increase oxide reliability.



Look out for **repetitions**. In 8, you can remove three instances of the word *category*.

8a. There are three categories of decentral heating. The first category is... The second category is... The third category is...



8b. The three main categories of decentral heating are... , ... , and



Check if you can turn **nouns into verbs**. You can often recognize such nouns by the word *of*, as in 9. The verb is usually more concise than the corresponding noun, and clearer.

9. **NOUN**

Removal of the rust was accomplished...
Increased warmth is an indication of...
We performed an analysis of...



VERB

The rust was removed...
Increased warmth indicates...
We analyzed...



Summary Principle 5: Concise

The principle Concise says that you should say what you want to say in the minimum number of words.

When a section or chapter of your text is almost ready, it is time to edit. Then remind yourself: **U SPArC**.